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Explaining destinations and volumes of international arms transfers: A novel network Heckman selection model

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ABSTRACT

What determines the volumes of international weapon transfers? And why do countries establish such arms trading relationships in the first place? We propose an innovative statistical strategy that builds on the gravity approach and combines a Heckman model with a network analysis. This allows us, for the first time, to analyze the impact of network structures on both the extensive and the intensive margins of the international arms trade simultaneously. We argue that the structure of the arms transfer network conveys important information for exporting and importing countries. Therefore, past topological properties of the trade network play a central role in its future evolution. Using data on the trade of major conventional weapons between 1955 and 2018, our estimation results and out-of-sample predictions show that network structures have considerable explanatory power with respect to the creation of trade links. They are far less relevant for the explanation of trade volumes, which are mainly determined by demand factors.

1. Introduction

According to data collected by the Stockholm International Peace Research Institute (SIPRI), international transfers of major conventional weapons (MCW) have continued to grow in recent years. Trade volumes have increased by 7.8 per cent between the five year periods of 2009–13 and 2014–18, thus reaching the highest level in the post-Cold War era (SIPRI, 2019, 8). While the international arms trade, henceforth IAT, is only a small part of the overall international trade in goods and services, it still warrants academic (and political) scrutiny, for it has not only economic consequences but also profound effects on the distribution of military capabilities, the structure of international relations as well as questions of war and peace. What determines the volumes of weapon transfers? Why do arms-producing countries trade arms with certain states but not with others? Research on the IAT has seen a rising number of contributions in recent years (e.g. Krause, 1995; Brzoska, 2004; Childs, 2012; Akerman and Seim, 2014; Martinez-Zarzoso and Johannsen, 2019; Thurner et al., 2019; Lebacher et al., 2020; Beardsley et al., 2020; Blum, 2019; Mehrl and Thurner, 2020) but there are still considerable theoretical and empirical gaps.

Since Tinbergen (1962), the standard framework to empirically analyze international trade flows has been the gravity equation, which takes into account economic mass and trade barriers due to geographic and social distance (for an overview, see Anderson and Van Wincoop, 2004; Anderson, 2011; Baltagi et al., 2017). However, the transfer of weapons differs substantially from trade in other goods and services. First, governments play a much more critical role. Even if manufacturers are private companies, they typically require explicit government permission for every export. When analyzing the IAT, it makes therefore sense to look at states

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as central actors. Second, trading arms has important security and political implications for a country, making their transfer much more sensitive.

A few studies have focused on individual countries and their export behavior (Blanton, 2000, 2005; Perkins and Neumayer, 2010; Schulze et al., 2017). However, a different strand of research has proposed that the IAT should be conceived of as a complex network, where dyadic trade relationships are dependent on the surrounding network and should therefore not be analyzed in isolation (see, for instance, Thurner et al., 2019; Lebacher et al., 2020).

While research utilizing a network perspective has been steadily growing in the social sciences,[1] taking a network perspective in analyzing international arms flows entails considerable methodological challenges. One statistical solution to account for complex network dependencies is the temporal exponential random graph model (tERGM, see Leifeld et al. (2018)) implemented by Thurner et al. (2019). Yet, existing studies either only examined the binary choice to transfer weapons (Thurner et al., 2019) or the determinants of trade volumes (Lebacher et al., 2020). However, as Helpman et al. (2008) have argued, trade consists of both an extensive margin, i.e., deciding with whom to trade, and an intensive margin, i.e. how much to trade. These two interdependent decisions can be thought of as a two-stage political process, where first trading partners are selected, and second, trade volumes are determined. We are aware of only one empirical study by Martinez-Zarzoso and Johannsen (2019) that analyzes both stages as interdependent decisions. However, their analysis ignores the influence of network structures on trade choices and may therefore be theoretically and empirically incomplete.

The aim of this paper is twofold. First, we take into account the insights of the theoretical network literature on trade (for overviews, see Rauch, 1999; Kranton and Minehart, 2001; Jackson, 2008) and apply them to the IAT. We propose two theoretical mechanisms that explain how network structures shape states' behavior by (a) affecting the costs of creating trade links and (b), more specifically, by reducing informational barriers. Second, we conduct an empirical analysis that, for the first time, analyzes both the intensive and the extensive margins of trade simultaneously while incorporating the pre-existing relational structures of the international arms trade network. Unfortunately, a TERGM cannot be applied in this situation because the inclusion of valued network structures in the second stage is conceptually not feasible and simultaneous estimation of network dependencies not possible. We, therefore, propose a different approach where we account for binary network dependence structures in both stages. As we assume that governments observe past transfer behavior when making export decisions, these network dependencies are first-order lagged. This conceptualization allows the application of a joint Maximum Likelihood selection model that includes relational and structural covariates. Thus, we can for the first time test the impact of network structures on both the extensive as well as the intensive margins and control for the fact that trade partners are not selected randomly.

We empirically test the effect of network structures based on a data set that is split into a Cold War (1955–1991) and a post-Cold War (1992–2018) sample in order to account for potential temporal heterogeneity. Our network Heckman selection model consists of two gravity equations – one for the choice of trade partners and one for trade volumes. In addition, we also conduct for both estimation stages in-sample and out-of-sample prediction analyses. This allows us to compare the model performances of specifications with and without network variables. Our overall conclusion is that network variables are important and that informational barriers as well as the costs of creating trade links strongly affect the choice of trade partners. They are less relevant in determining trade volumes where demand factors play a more prominent role.

The paper is structured as follows. The next section explains our theoretical arguments in more detail. This is followed by a discussion of our empirical approach. Section 4 describes our data and provides some descriptive statistics, while section 5 presents our estimations results and compares out-of-sample model performances. Finally, we provide a summary and discuss our conclusions.

2. The international arms trade as a complex network

2.1. Costly links and the role of informational barriers

Past research has highlighted the importance of economic incentives but also strategic considerations in the IAT (Levine et al., 1994; Levine and Smith, 1997). States don't want to deliver arms to clients that could pose a future threat to them or their allies. Security risks but also the perception of benefits may be linked to factors such as domestic politics (Comola, 2012), regime type (Martinez-Zarzoso and Johannsen, 2019), position in the international trade network, for example for oil (Bove et al., 2018), as well as the affinities of suppliers and clients to other competitors in the international system (Harkavy, 1994; Krause, 1995).

A parsimonious theoretical approach that includes both economic and security interests in a country's decision-making process was developed by Levine et al. (1994). This formal model was later extended to include domestic arms consumption to analyze arms races (Levine and Smith, 1995; Garcia-Alonso and Levine, 2007) and the interaction between exports and military expenditures (Pamp et al., 2018). The welfare function of an exporting country can be written as

$$W_t = U(S_t) + p(x_t)x_t - C(x_t) \tag{1}$$

where S_t captures the security effect of exporting arms, $p(x_t)x_t$ is the revenue from exporting x_t units of weapons, and $C(x_t)$ denotes the production cost function. This approach is focused on the analysis of aggregate trade volumes but less well suited to explain

[1] For an introduction with an international relations focus see Maoz (2011), for a comprehensive recent overview see Victor et al. (2017). Examples of specific topics are alliances and international cooperation (Maoz and Joyce, 2016; Warren, 2016), conflicts and their resolutions (Böhmelt, 2016; Ward and Dorussen, 2016; Wilson et al., 2016) and global oil trade networks and military spending (Bove et al., 2016).

binary export decisions. What is missing is the strategic, political as well as informational costs-benefits calculus of engaging in an arms transfer. Especially the cost perspective has been neglected, despite the fact that it could be crucial in explaining a seller *i*'s decision to export to buyer *j* instead of buyer *k*.

Establishing an arms trade relationship is costly. Economic costs arise from the need to facilitate the transaction (i.e. costs of surveying the market, initiating, forming and monitoring an agreement) or from possible concessions one country has to make in other areas – for example allowing favourable access to domestic markets for other goods. Furthermore, there are risks associated with payment default, property rights, unlicensed re-selling of arms and so forth.

There are also important security costs that result from the fact that military capabilities are transferred, thus increasing the military power of the buyer. Hence, the reliability of the arms receiver becomes a crucial issue. While this may be less problematic in the case of a close ally, the less aligned two states are in foreign policy terms, the higher the potential security costs.

Finally, establishing links with other states could entail political costs, both domestic and foreign. Exporting (or importing) arms to (from) certain countries could antagonize other states, which may then lead to further economic and security costs. Creating trade links with certain states may prove unpopular domestically, thus affecting the electoral prospects of a government. Hence, $C(x_t)$ should be conceived of as not only including production costs but these other costs of establishing a trade link as well. Incomplete information about the costs is thus an important impediment to trade. Moreover, the precise effect of an arms transfer on security S_t may not be fully known in advance, thus constituting an additional source of uncertainty.

The network perspective has long emphasized the importance of informational barriers in preventing the creation of trade relationships (Pomeroy, 1984; Rauch, 1999). Knowledge about market conditions, demand, regulatory frameworks, reliability and trustworthiness of the buyer are but a few factors that may prevent a seller from expanding into new locations abroad. Chaney (2014) argues that a central role of networks is to help to overcome these informational frictions. For example, a firm that has traded with *k* will be more inclined to establish a link with *j* if *k* and *j* have previously traded with each other (Chaney, 2014, 3601). Hence, the network itself provides valuable information that helps firms to establish new links.

The theoretical network literature (for overviews, see Jackson, 2008; Fagiolo, 2016) has so far exclusively focused on firm-level or individual-level behavior (e.g. Rauch, 2001; Di Gioacchino and Fichera, 2020), but it can easily be transferred to the strategic behavior of countries engaged in arms trades.[2] We follow (Thurner et al., 2019, 9) and formally describe the arms trade network as consisting of a set of countries $S_t = 1, 2, \ldots, N_t$ and a matrix X_t with cell entries $x_{ij,t} \geq 0$ capturing the sum of exports from i to j in period t. Both the set of countries and export sums can vary over time with $t \in \{t_0, \ldots, t_k\}$. Note that we go beyond Thurner et al. (2019) who only analyze whether trade relations exist. A major contribution of our approach is that we consider both the creation of trading links and the volumes being transferred.

Trading requires countries to invest in establishing and sustaining links (Uzzi, 1996). The link costs have to be weighed against the economic and security benefits of trading arms. However, informational barriers prevent countries from knowing the precise economic consequences or political and security repercussions of weapon transfers. Thus, link costs and ex-ante uncertainty about them substantially affect the IAT. We assume that countries are heterogeneous and link costs are not perfectly known in advance. Our main argument is that the network structure itself affects the size of these costs and mitigates informational barriers. We propose two theoretical mechanisms: 'cost reduction' and 'information dissemination'. First, past investment in links reduces the costs of establishing new links. Countries with many existing trade relationships should find it easier to establish new ones. Second, existing links can convey useful information about costs and therefore reduce informational barriers. Countries engaged in many trade relationships reveal more information about themselves. In the simplest case, an exporter i may observe a trade link between importer j and another exporter k. This reveals useful information about link costs.

Different topological structures relate to cost reductions and information dissemination. The simplest form are dyadic relationships established in the past between an exporter i and an importer j. The costs have already been borne in $t-1$ and information about it has been revealed. While the costs of sustaining a link may change and may not be fully known, an existing relationship should, ceteris paribus, make future trade more likely. Furthermore, countries that transfer arms with each other usually have compatible arms systems. An example is the strategy of the EU to strengthen further its common defence technological and industrial base (DTIB), which is expected to even further increase intra-EU arms trade due to more similar arms systems (see Chagnaud et al. (2015)). We, therefore, expect reciprocity to make weapon transfers more likely.

Another important issue arises with respect to the costs of severing a link. The network literature usually assumes that un-linking is without costs. While this might be a reasonable assumption for many other types of networks, it could be problematic in the case of the IAT. Ending an arms trade relationship may have not only economic consequences but also severe security repercussions. For example, an exporter's decision to stop weapon deliveries forces an importing country to look for new suppliers, which may cause a foreign policy realignment of the importer. Moreover, the supplier may have to shift the export of already manufactured products to other countries. Hence, if ending trade ties is costly, we expect past links to be even more relevant for future trade relations. Prominent examples are the long-standing ties of the US with India or with Saudi Arabia. The US can rely on established relationships with these countries, which are dependent on arms imports as they have no sizable domestic production (see Chiriyankandath, 2004; Bronson, 2008).

The number of export links is another important network property. Having established many export ties in the past should lower the costs of creating new links. Furthermore, past experience should reduce informational barriers regarding the costs of future ties. The higher the number of ties, the easier it is to create new ties. This dynamic helps to explain the emergence of so-called super-

[2] We neglect the possibility of illegal transfers and therefore ignore the international black market in arms because it is beyond our theoretical and empirical scope.

sellers like the US or Russia. Note that we can also analyze the number of import links of buyer j. The theoretical rationale is that the existence of many ties signals solvency and trustworthiness and therefore reduces costs and informational barriers to creating new links.

There are also hyper-dyadic connections. For example, if in $t-1$ country i exports to a third country k, which in turn delivers arms to j, then the likelihood of exports from i to j should increase. This could best be described as 'customer-of-my-customer' relationship. The existing link between k and i reduces link costs for a tie between i and j. In particular, if i and k share a similar weapon technology, then transfers by k to j should facilitate exports by i because of the compatibility with j's arms. Moreover, observing the existing link should reveal important information about the costs for i of establishing a link with j. This is, of course, not restricted to only one third country but can be generalized to a higher number of k-countries. Typical examples are medium-sized exporters such as the European countries who often transfer arms to each other but also share ties to large buyers such as Egypt, Turkey and Saudi Arabia (Fleurant et al., 2016).

Similarly, there could be shared supplier or shared customer relationships. If, for instance, both i and j imported from (or exported to) the same country k in $t - 1$, then we expect there to be a higher chance of a trade between the two in period t. The existence of shared supplier or shared customer ties reduces the cost of creating a new link between i and j and reveals information about the size of the costs. This is reinforced with each additional shared supplier. A prominent empirical example of these types of relationships is the US which is a major supplier to France, the UK, Germany and Saudia Arabia as well as Israel, who in turn trade heavily with each other. As a non-western example, Russia was (and still is) a shared supplier for China and Iran in the early post-Cold War years (see Harkavy, 1994).

In sum, we argue that the very structure of the arms network influences trade by affecting the costs of establishing links and reducing informational barriers. Dyadic and hyper-dyadic relationships reduce uncertainty about economic, security and political costs. We, therefore, expect empirically that a weapon export from i to j becomes more likely if there exists a prior arms trade relationship between the two. Furthermore, we expect a positive effect of the number of dyadic links that exporting and importing countries have (i.e. their outdegrees and indegrees). Finally, trade between two countries should be positively affected by the number of in-direct (i.e. hyper-dyadic) relationships they have.

2.2. The extensive and intensive margins of the IAT

So far, we have discussed the IAT in very general terms as a question of whether or not any transactions take place. In their seminal paper, Helpman et al. (2008) have argued that when analyzing trade frictions, one should analytically decompose trade flows into the extensive and intensive margin. This distinction is empirically necessary but also encapsulates a government's decision-making process, which consists of two stages. From an exporting country's perspective, stage one is the decision of whether an arms export to another state should be granted; at stage two, trade volumes are determined.

Our theoretical arguments about the effect of the arms trade network's topology have been developed with respect to the first decision stage. But what about their impact on trade volumes? If link costs and informational barriers are proportional to trade volumes, then we would expect that network properties have a similar effect on both margins. If link costs and information barriers are fixed, then they should be less relevant with respect to the intensive margin. Furthermore, supply constraints and demand factors could play a much more important role. Limited capacity to expand production would mean that a rising number of export links may not entail increasing trade volumes – indeed, it could even imply smaller volumes. The number of import links could be an indicator of the intensity of demand that has nothing to do with informational frictions or link costs.

In sum, the effect of cost reduction and information dissemination is less clear-cut for the intensive margin than the extensive margin. Therefore, empirically, we expect a positive effect of the dyadic and hyper-dyadic network variables on trade volumes only if link costs and informational barriers are proportional to the size of arms exports. However, if link costs and informational barriers are fixed, we would expect no significant effect of these variables on trade volumes. Moreover, if supply constraints and demand patterns are important, we expect some network variables to have a negative effect.

3. Empirical strategy: combining a Heckman selection model with a network gravity equation

A major caveat when analysing network data comes with the definition of network dependence structures. In our example, an important network dependency could be, for example, reciprocity, i.e. the tendency of countries to be reciprocal in their arms trade relationships. The literature on statistical network data analysis has proposed different approaches to taking these dependencies into account. One possibility is to use a stochastic actor-oriented model (SAOM) that regards the process of network building between two time points as a continuously evolving process where ties are allowed to be added and removed during the two observed points in time (see Kinne, 2016). While this might be appropriate for friendship networks, it is not for arms trade networks where a tie indicates the existence or non-existence of a trade relationship in a given year. Hence, this mechanism does not fit into the SAOM method that requires that these ties can disappear and reappear again between two time points.

Another option, as proposed by Thurner et al. (2019), is the use of some kind of temporal ERGM. In this model class, network dependencies can be modelled simultaneously but most often they are incorporated through their impact over time. We argue that in the given case, this is the more appropriate way of dealing with network dependencies because arms trade often comes with a lag between the ordering and the delivery of the weapons. Using the example of reciprocity again, it appears to be more plausible that a transfer from country i to j in $t-1$ is reciprocated by country j in t than both trades happening at the same time. This reasoning is also in line with the findings of Lebacher et al. (2021). However, the question of how to handle large scale weighted networks in a consistent way in the ERGM framework remains unresolved. We, therefore, propose to use a Heckman selection model (Heckman, 1977) as a substitute where we specify the binary model component (the probit model) similar to a temporal ERGM and additionally enrich the second stage of the model, the regression part, with network variables. By doing so, we are able to model for the first time the binary relations in a tERGM-like fashion while also considering the trade volumes in a consistent framework.

Employing a Heckman selection model also makes theoretical sense. As Helpman et al. (2008) have demonstrated, if trade decisions follow a two-stage decision process, where first trading partners are selected before trade volumes are determined, then a zero volume transfer should be interpreted as a latent outcome of the decision not to trade with a certain country. Since these two stages are potentially not independent from one another, ignoring the first stage – the selection stage – would introduce bias at the second stage, i.e. the estimated trade volumes. This should be particularly severe for the IAT because export decisions by countries are made not only based on economic criteria but also based on political and security considerations. Furthermore, arms transfers are relatively rare compared to general goods trade relations. Thus, for most dyad-years we find zero transfers. As a result, a two-stage estimation approach is needed, which takes the interdependence of the two stages into account. Note that a two-stage approach is also warranted from a network perspective. Research on trade networks (see Fagiolo et al. (2008)) comes to the clear conclusion that the determinants of both binary and valued edges need to be analyzed.

Formally, the model we estimate[3] can be written as

$$Y^{\star}_{ij,t} = Z'_{ij,t-1}\gamma + u_{ij,t} \tag{2}$$

$$D_{ij,t} = 1(Y^{\star}_{ij,t} > 0) \tag{3}$$

$$Y_{ij,t} = X'_{ij,t-1}\beta + \epsilon_{ij,t} \tag{4}$$

$$\begin{pmatrix} u_{ij,t} \\ \epsilon_{ij,t} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & \sigma^2_{\epsilon} \end{pmatrix}\right), \tag{5}$$

where $Z'_{ij,t-1}$ and $X'_{ij,t-1}$ are our lagged explanatory variables. We consider dyadic arms trade volumes among n_t countries with $i, j \in S_t$, where S_t refers to the set of countries in t. At the first stage, we have the realization of the latent variable $Y^{\star}_{ij,t}$. We observe arms transfers only if $Y^{\star}_{ij,t} > 0$. As an indicator for whether a trade takes place, we define $D_{ij,t}$, which is one if $Y^{\star}_{ij,t} > 0$, and zero otherwise. The second stage takes into account all cases where trades are realized ($D_{ij,t} = 1$), with the dependent variable $Y_{ij,t}$ being the logged trade volume.

This model can be estimated using Heckman's two-step approach, which combines the probit estimator of the first stage with OLS at the second stage. However, for example Nawata (1994) has shown, Maximum Likelihood (ML) is much more efficient, especially when there is a high degree of multicollinearity between the covariates and the hazard ratio. Our model is therefore estimated by joint ML.[4]

A critical question arises with respect to the exclusion restrictions. In principle, the model above can be estimated without an exclusion restriction. This, however, shifts the whole identification strategy to the parametric distributional assumptions. In order to circumvent this, we follow the standard practice and rely on an instrument for identification. Formally, the model is fully identified if $Z_{ij,t}$ contains at least one component that (a) has explanatory power for the selection stage and (b) does not contribute to the second stage (i.e. is a valid instrument). We choose a dummy variable that captures whether a receiver country j experiences any type of violent conflict, which we found significantly raises the likelihood of trade flows but not trade volumes. We also tested alternative exclusion restrictions. For example, in accordance with Martinez-Zarzoso and Johannsen (2019), we tried a dummy variable that indicates whether an importing country is under an arms embargo. This choice did not affect our results with respect to the network variables. However, we decided against using this exclusion restriction because it turned out that the embargo variable is not significant at the first stage for the Cold War period. This is not surprising because embargoes have been repeatedly violated during this period (Erickson, 2013; Schulze et al., 2017). In general, we found that our results proved to be robust to the choice of the exclusion restriction. We are very confident that other suitable exclusion restrictions would lead to similar results.

We split our sample into two periods: the Cold War (1955–1991) and the post-Cold War era (1992–2018).[5] Past research clearly shows strong qualitative differences between the two periods (Brzoska, 2004; Thurner et al., 2019). The IAT has become much more commercialized and less segmented along the East-West divide. Even though this trend had already started before 1991, the end of the Cold War accelerated these processes and therefore represents a clear structural break.

The decision to lag all independent variables by one year follows from our theoretical argument that network structures affect the future evolution of the IAT network. Using lags for estimating an evolving network, we follow the approach by Franzese et al. (2012), which demonstrates the advantages of including lagged variables in network models.

Lagged variables are not the only way we model temporal effects. We estimated both stages using either time polynomials up to degree three (see Carter and Signorino, 2010) or time fixed effects. Another option would be to include country fixed effects. This

[3] A more in-depth discussion of our estimation strategy can be found in the Online Appendix.

[4] Note that we also tested all our models using the two-step estimator. However, coefficients and statistical significance levels were close to being identical.

[5] We chose 1991 as a cutoff because it marks the end of the Soviet Union.

could be one way to control for persistent differences in domestic arms production capacities. However, we found that controlling for GDP and domestic military expenditures captures differences in production capacities sufficiently well. Therefore, we decided against fixed effects to avoid the resulting massive increase in the number of parameters that would have to be estimated. However, as a robustness check, we also tested the Poisson Pseudo Maximum-Likelihood (PPML) estimator as proposed by Santos Silva and Tenreyro (2006) and implemented in the context of arms trade research by, for example, Bove et al. (2018). This approach permits us to include time-, importer- and exporter-fixed-effects. The results using this estimator were not too different to our second stage results but produced higher significance levels for a number of variables.[6] We prefer the Heckman joint ML estimator, as it proved to be the more conservative approach and, more importantly, allows us to directly test whether network effects have different impacts on the extensive and intensive margins of trade.

Finally, our estimations are based on a network version of the 'gravity of trade' approach. We, therefore, take the importance of the trading partners' economic sizes and their spatial as well as political distances into account. Indeed, network analysis is particularly well suited to control for these factors because it allows specifying nodal as well as relational attributes. As a result, we arrive at a computationally very demanding, two-stage selection network model that allows joint estimation of both the extensive and the intensive margins. To our knowledge, we are the first to apply such an estimation approach to the empirical study of the international arms trade.

4. Data description

4.1. The dependent variable

Our empirical analysis focuses on major conventional weapons. Data is provided by the Stockholm International Peace Research Institute and contains information on all transfers of MCW from 1955 to 2018. The data on trade flows includes information on the sender country, the receiver country as well as the type of weapon that is traded, which include, for example, aircraft, armoured vehicles and ships.[7] SIPRI records order and delivery dates but the order dates are far less reliable, and we, therefore, use delivery dates for our analyses. Furthermore, using delivery dates accounts for the fact that arms are often split into several installments, and deliveries are easier observed by other actors in the network than orders. Of course, in using delivery dates, we cannot account for orders that are not being fulfilled for various domestic and international reasons.[8] However, in line with our theoretical approach this would mean that the creation of a link turned out to be too costly. Using order dates in such a situation would mean to erroneously measure a link that actually has not been realized.

The volume of the trades is measured in so-called TIV – shorthand for trend-indicator value, which is expressed in millions. These values are not the sales prices of arms transfers but are based on the estimated production costs of core weapon systems. They represent the amount of military resources that are traded and can be easily compared across years.[9]

Fig. 1 depicts the IAT networks for the years 1950, 1980, 1990 and 2017 and therefore provides a description of the dynamics of the extensive margin. Note that the size of the nodes increases with export volumes. A picture emerges of a small set of big exporters and a large number of smaller importing countries. While the network has become denser over time, indicating the dissolution of the East-West divide and globalization of the IAT, certain export clusters have persisted. Clearly, arms transfer relations have changed considerably over time, both with respect to the extensive as well as the intensive margins. The clustering of trade activity throughout the whole period around a number of central actors such as United States (USA), Soviet Union (RUS)/Russia (RUS), China (CHN), France (FRA), Germany (DEU) and United Kingdom (GBR) remains a prominent feature.

Based on the dyadic data set from SIPRI, we constructed a sequence of weighted adjacency matrices A_t where t denotes the respective years ranging from 1955 to 2018. Within these adjacency matrices, an entry $X_{t,ij}$ corresponds to the trade volume exported from country i to country j in the year t. Additionally, we enrich the data by dyadic and monadic covariates.

4.2. The independent variables

All independent variables are lagged by one year[10]. The variable 'past link' is a dummy that indicates whether a transfer from i to j took place in $t-1$. Conversely, 'reciprocity' is also a dummy and measures if a transfer from j to i took place in $t-1$. The variable '# of exporting links (i)' indicates the number of existing export partners and is operationalized by $(n_{t-1}-1)^{-1}\sum_k Y_{ik,t-1}$, which is the number of outdegrees of i divided by the total number of theoretically possible outdegrees i could have had in $t-1$. Note that we also include the number of exporting links of importer j ('# of exporting links (j)'). Furthermore, '# of importing links (j)' measures the number of import links and is calculated by $(n_{t-1}-1)^{-1}\sum_k Y_{kj,t-1}$, which represents the number of j's indegrees as a share of the total number of possible indegrees in $t-1$. We also include the import links of the exporting country ('# of importing links (i)').

[6] The higher significance can in part be attributed to the fact that PPML uses many more observations because there is no selection stage. It also overestimates, in our view, the importance of past trade relationships for export volumes. Overall, the PPML estimator finds more significant effects of network variables on export volumes than our Heckman estimator. These results can be found in the Online Appendix.

[7] See Table 6 in the Online Appendix for a comprehensive overview of the types of arms included and Table 7 for the list of countries included.

[8] For instance, a change of government, domestic resistance or international reputational cost may all lead a government to not fullfil an order.

[9] For detailed explanations on the methodology, see SIPRI (2017b).

[10] The decision to use a one-year lag instead of the often used 5-year lags was taken in order to not 'over-smooth' the results.

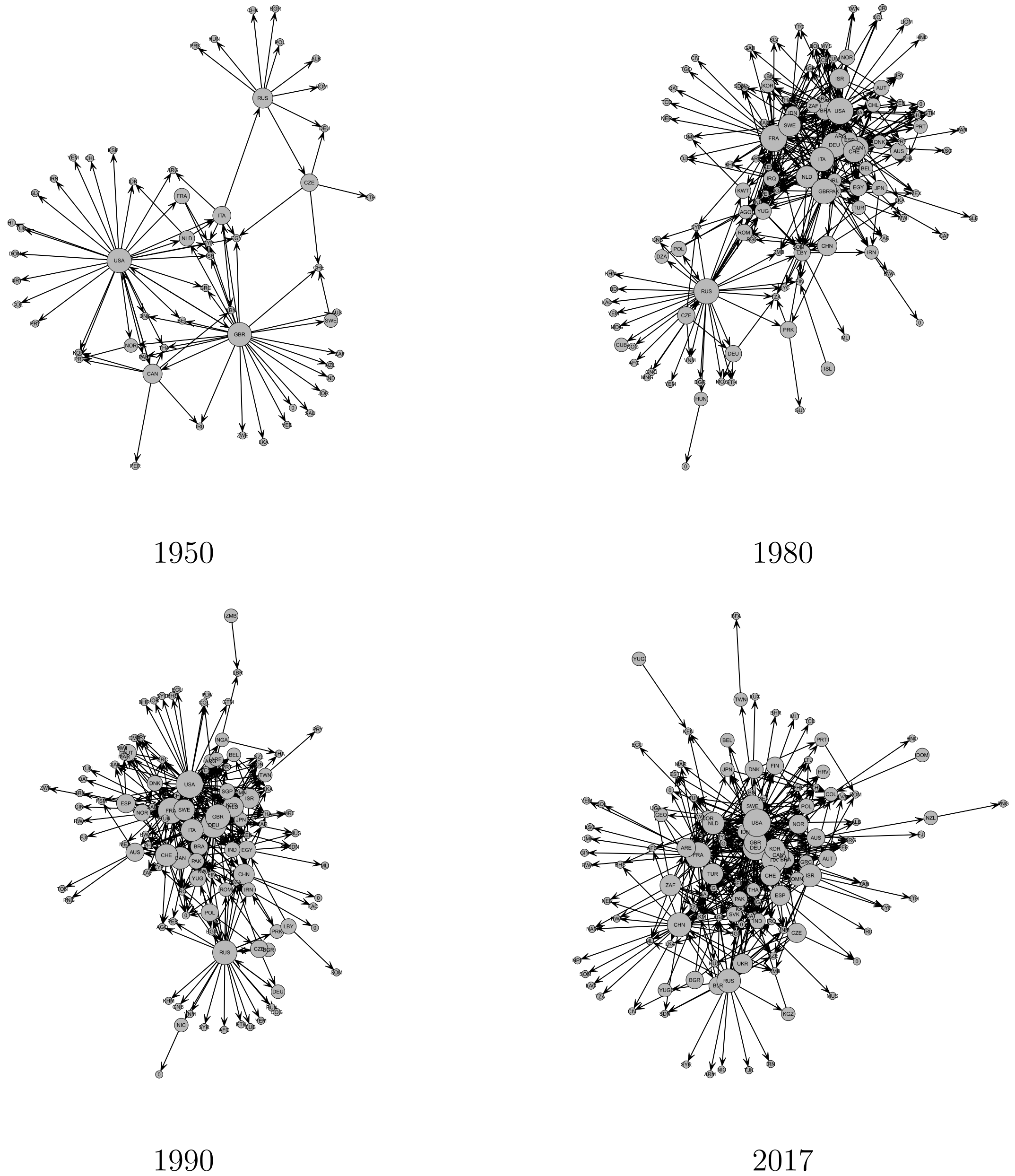


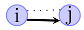
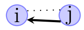
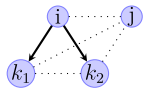
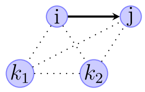
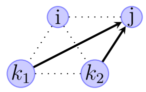
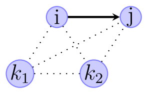
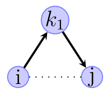
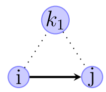
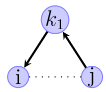
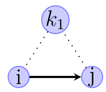
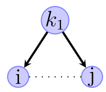
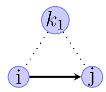
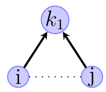
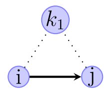
Fig. 1. International arms trade network in selected years.

Regarding the hyper-dyadic variables, we include 'exporting transitivity' which measures the number of indirect export ties relating i to j via connecting actors k divided by all the indirect ties i could have had to j in $t-1$: $(n_{t-1}-2)^{-1}\sum_k Y_{ik,t-1}Y_{kj,t-1}$. Conversely, 'importing cycle' counts the number of indirect export ties from j to i via connecting actors k divided by all the indirect ties j could have had to i in $t-1$ ($(n_{t-1}-2)^{-1}\sum_k Y_{jk,t-1}Y_{ki,t-1}$). We measure 'shared supplier' relationships by $(n_{t-1}-2)^{-1}\sum_k Y_{kj,t-1}Y_{ki,t-1}$, which counts in $t-1$ the number of import ties i and j have with third countries k divided by the total number of possible shared import links. Finally, 'shared customer' captures in $t-1$ the number of export ties i and j have with third countries k divided by the number of all possible shared export ties: $(n_{t-1}-2)^{-1}\sum_k Y_{ik,t-1}Y_{jk,t-1}$.

Table 1 provides a summary of our network variables and shows how these different network structures in $t-1$ affect trade relations between exporter i and importer j in period t. It also shows how we formalize these concepts. Note that we relate the number of observed ties to the total number of ties that are numerically possible.

We include a number of covariates that have been found to be important in past research on the demand for arms imports (Smith and Tasiran, 2005, 2010) and the arms trade network (Kinne, 2016; Thurner et al., 2019). With respect to gravity variables, we

Table 1
Network variables in $t-1$ and their effect on the links between exporter i and importer j in t.

Network Structures	$t-1$	t
past link $Y_{ij,t-1}$ (binary)		
reciprocity $Y_{ji,t-1}$ (binary)		
# exporting links (i) $(n_{t-1}-1)^{-1}\sum_k Y_{ik,t-1}$		
# importing links (j) $(n_{t-1}-1)^{-1}\sum_k Y_{kj,t-1}$		
exporting transitivity $(n_{t-1}-2)^{-1}\sum_k Y_{ik,t-1}Y_{kj,t-1}$		
importing cylce $(n_{t-1}-2)^{-1}\sum_k Y_{jk,t-1}Y_{ki,t-1}$		
shared supplier $(n_{t-1}-2)^{-1}\sum_k Y_{kj,t-1}Y_{ki,t-1}$		
shared customer $(n_{t-1}-2)^{-1}\sum_k Y_{ik,t-1}Y_{jk,t-1}$		

include the logarithmic distance between the capital cities of countries i and j ('log distance')[11] which is independent of time (with the exception of the case of Germany which changed its capital after the reunion). Trade theory predicts a negative impact of distance on the probability and volumes of arms transfers.

Similarly, the closer two countries are politically, the more likely they should be to engage in trade relations. We, therefore, include the distance between two political regimes measured as the absolute differences between their Polity-IV-scores. The regime scores range from −10 (hereditary monarchy) to + 10 (consolidated democracy) and are explained in Marshall and Jaggers (2002).

Belonging to the same alliance could make trade more likely and may also affect volumes. We therefore construct a dummy variable called 'alliance', which is based on the data provided by the 'Alliance Treaty Obligations and Provisions Project' (see Leeds et al., 2002). It takes the value one if i and j are part of a formal alliance, which can take the form of a defence and offensive support, neutrality, non-aggression or consultation pact.[12] Both, Polity IV scores and alliances are time dependent and can be represented by adjacency matrices.

Additionally, we include a number of monadic (country-specific) variables that are time-dependent but do not require a representation as an adjacency matrix. We control for the economic masses of both the exporting ('log GDP (i)') and importing country ('log GDP (j)') by taking the log of the real GDP based on data from Gleditsch (2013) and the World Bank. Similarly, we also add 'log population (i)' and 'log population (j)' to capture population sizes.[13]

We expect arms flows to be closely related to a country's military expenditures. Indeed, most studies have neglected this factor so far. The size of defense budgets could be a proxy for a country's military capacity as well as its home market size and thus its ability to produce (and therefore export) weapons. It could also indicate a country's willingness to commit resources to the military and thus the demand for weapons. We therefore include logged military expenditures for both exporter ('log military expenditure (i)') and importer ('log military expenditure (j)'). We decided to use absolute values measured in constant US dollars using purchasing power parities. The data is from Nordhaus et al. (2012), who merge SIPRI data with information from the Correlates of War project

[11] The data is taken from Gleditsch (2013).

[12] Since the alliance data are available only up to the year 2016, we extrapolated the data to 2017. Note that we do not need data for 2018 because all our covariates are lagged by one year.

[13] Population and GDP data for the period 1955–2011 are from Gleditsch (2013), while the years 2012–2017 are from the World Bank's Development Indicators (World Bank, 2018). We recalculated the World Bank's GDP data by changing the base year to 2005 in order to make it compatible with the Gleditsch data. Note that estimating the post-Cold War sample using only the World Bank GDP data does not change our results but entails a much higher number of missings.

(see Singer et al. (1972)). We extended their data to 2018 using the most recent SIPRI (2017a) military expenditure data.[14]

Two further controls are 'embargo (j)' and 'path dependence'. The former variable is a dummy indicating whether an importer is under a multilateral arms embargo.[15] Data on embargoes are taken from SIPRI (2017c). The latter variable is a moving five-year average of exports (for the previous five years) from i to j to account for persistence in trading relationships. Furthermore, it also captures the fact that deliveries are often made in several installments over an extended period of time.

Finally, we add a variable named 'conflict' as our exclusion restriction. This is a dummy variable which indicates whether an importing country is involved in an inter-state, intra-state or extra-systemic violent conflict. The data is from the UCDP/PRIO Armed Conflict Dataset version 17.2 (see Gleditsch et al., 2002; Melander et al., 2016). A conflict is defined as a clash of armed forces that results in at least 25 battle-related deaths in a calendar year.

5. Results of the empirical analysis

5.1. First stage estimation results

Table 2 presents our estimation results for the first stage of the network Heckman selection model. Models (1) and (2) show the estimates for the Cold War era, whereas models (3) and (4) cover the post-Cold War period. Odd-numbered columns include time polynomials, even-numbered columns include time fixed effects. One striking feature is the robustness of our results to the type of time effect. Coefficient sizes and significance levels are almost always identical.

The upper half of the table shows the coefficients of our network variables. Most of the coefficients are in line with our theoretical expectations. Having established a trade link in the past strongly increases the probability of a trade relationship in the future. This effect is found for both sample periods. This finding is compatible not only with the argument that the costs of establishing a link have already been borne, but also that un-linking is costly and therefore exporters do not switch easily between buyers. Reciprocity is also highly significant and has the expected positive effect. If i has imported from j in the past, then exports from i to j become more likely. Again, investment in a link has already been made, and link costs revealed. This clear, statistically significant impact was not found in Thurner et al. (2019). The difference can probably be explained by the fact that we include military expenditures as a suitable proxy to control for production capacities. Note that our results indicate that the influence of reciprocity is smaller and slightly less significant for the post-Cold War period, which probably is due to the disintegration of the Soviet bloc and a re-orientation of many countries to new buyers and sellers.

The number of previous export links of a selling country has a positive and significant effect in all four models, with coefficients being higher for the period after 1991. This supports our theoretical notion that due to their past experience, exporters with many ties find it less costly to establish new trading links and face lower informational barriers. We find the opposite effect regarding the number of export links of importing countries, with coefficients being negative but of similar sizes. This result is not a surprise, however, since it merely captures the fact that countries with many export relationships necessarily have a sizable domestic arms production and are therefore less reliant on imports. This also explains why the number of importing links of i has a significantly negative effect on the likelihood of exporting to j.

The number of import links of importer j has no robust, significant effect on the likelihood of receiving an arms transfer from i. Theoretically, j having many import links should reveal more information about the cost of establishing an export tie with j, thus making trade easier. However, due to extreme market concentration in supply, we would expect (and we see this in the indegree distribution) that importers only have very few providers. This secures domestic interoperability of imported weapons. Also, there are strong pressures from exporters to not seek trade links with other exporters in order to prevent knowledge about proprietary weapons technology to be shared with rival producers. Moreover, we believe that this variable also captures a saturation effect. The more sources for import j has, the smaller the need and ability to establish new trading relationships with other exporters. Given that we identified a strong persistence in trading relationships, a rising number of import links naturally makes new links less likely.

Turning to hyper-dyadic relationships next, we find that exporting transitivity has the expected positive effect, which is highly significant and robust across all models and sample periods. Exporting to k which in turn has an export link with j makes it easier for i to establish an export link with j. Observing the link between k and j reveals information about link costs and makes trade with j more likely. Interestingly, the same cannot be said for importing cycle. The coefficients are negative and significant for the Cold War period and become positive and insignificant after 1991. Hence, such a relationship does not increase the likelihood of a trade link. This could suggest that from an exporter's perspective, observing export links from j to k does not reveal much information about the costs of establishing an export link with j. This implies that link costs and informational barriers of export links are not the same as import links.

Shared supplier and shared customer relationships have the expected positive effect on export probabilities. Note that the coefficients for the former are larger and more significant. However, they are much higher during the Cold War, which should be no surprise given this period's much more polarized strategic environment. In sum, most of the network variables have the expected effects (with the exception of '# importing links (j)' and 'importing cycle'), suggesting that cost and information dissemination indeed

[14] We used linear interpolation to reduce the number of missings.

[15] Note that embargoes could be a leading indicator of future economic sanctions which have been shown to lead to increases in goods trading due to incentives for stockpiling (Kwaku Afesorgbor, 2019).

Table 2
First stage results.

	(1) 1955–1991 Time Poly.	(2) 1955–1991 Time FE	(3) 1992–2018 Time Poly.	(4) 1992–2018 Time FE
Network Statistics				
past link	1.45***	1.46***	1.47***	1.47***
	(0.02)	(0.02)	(0.02)	(0.02)
reciprocity	0.29***	0.29***	0.11**	0.10*
	(0.05)	(0.05)	(0.04)	(0.04)
# exporting links (i)	0.01***	0.01***	0.02***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)
# exporting links (j)	−0.01***	−0.01***	−0.02***	−0.02***
	(0.00)	(0.00)	(0.00)	(0.00)
# importing links (i)	−0.04***	−0.04***	−0.06***	−0.05***
	(0.01)	(0.01)	(0.00)	(0.00)
# importing links (j)	−0.01*	−0.01	0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.01)
exporting transitivity	0.11***	0.11***	0.10***	0.10***
	(0.01)	(0.01)	(0.01)	(0.01)
importing cycle	−0.06*	−0.06*	0.02	0.02
	(0.03)	(0.03)	(0.02)	(0.02)
shared supplier	0.17***	0.17***	0.08***	0.08***
	(0.01)	(0.01)	(0.01)	(0.01)
shared customer	0.01*	0.01*	0.01*	0.02*
	(0.01)	(0.01)	(0.01)	(0.01)
Controls				
path dependence	0.30***	0.30***	0.29***	0.29***
	(0.01)	(0.01)	(0.01)	(0.01)
alliance	0.13***	0.13***	−0.07***	−0.07***
	(0.02)	(0.02)	(0.02)	(0.02)
log GDP (i)	0.12***	0.12***	0.27***	0.27***
	(0.01)	(0.01)	(0.01)	(0.01)
log GDP (j)	0.02	0.02	−0.03**	−0.03**
	(0.01)	(0.01)	(0.01)	(0.01)
polity difference (ij)	−0.01***	−0.01***	−0.01***	−0.01***
	(0.00)	(0.00)	(0.00)	(0.00)
log military expenditure (i)	0.20***	0.19***	0.02*	0.02*
	(0.01)	(0.01)	(0.01)	(0.01)
log military expenditure (j)	0.00	0.00	0.08***	0.08***
	(0.01)	(0.01)	(0.01)	(0.01)
log population (i)	−0.06***	−0.06***	−0.12***	−0.13***
	(0.01)	(0.01)	(0.01)	(0.01)
log population (j)	0.03***	0.03***	0.05***	0.05***
	(0.01)	(0.01)	(0.01)	(0.01)
log distance (ij)	−0.17***	−0.17***	−0.21***	−0.21***
	(0.01)	(0.01)	(0.01)	(0.01)
conflict (j)	0.09***	0.09***	0.07***	0.07***
	(0.02)	(0.02)	(0.02)	(0.02)
embargo (j)	−0.02	−0.02	−0.10**	−0.10**
	(0.07)	(0.07)	(0.04)	(0.04)
AIC	65344.72	65345.20	70170.14	70175.48
BIC	66677.37	65938.04	71292.77	70768.28
Log Likelihood	−32553.60	−32619.36	−34988.74	−35032.07
N	537474	537474	588754	588754

Note: Intercept not shown. *i* is the exporting country and *j* the importing country. Standard errors in parentheses. Coefficients of time dummies not shown. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

affect the likelihood of arms exports.

Taking a brief look at the control variables reveals that the gravity variables have the expected coefficients. Spatial and political distances significantly reduce the chances of creating trade links in all models. The GDP of the exporter has a positive effect on export probabilities in both periods. This is due to the fact that the ability to produce arms domestically correlates strongly with economic development. Note that GDP of the importer is only statistically significant in the period after 1991, where the coefficient is negative. This underlines again that real GDP is a strong proxy for the ability to produce and export arms.

The results for the military expenditures variables confirm the increasing commercialization of the IAT. During the Cold War period, only the exporting countries' defense budget sizes matter, whereas the importers' expenditures are insignificant. This reflects the fact that a considerable part of arms transfers during that period came in the form of military assistance (Brzoska, 2004, 115).

Table 3
Average marginal probability effects when changing variable values to the fourth quantile, first stage.

	(1) 1955–1991 Time Poly.	(l) 1992–2018 Time Poly.
past link	0.077	0.089
reciprocity	0.005	0.002
# exporting links (i)	0.008	0.058
# exporting links (j)	−0.005	−0.010
# importing links (i)	−0.005	−0.011
# importing links (j)	−0.002	0.0001
exporting transitivity	0.027	0.084
importing cycle	−0.006	0.006
shared supplier	0.031	0.027
shared customer	0.007	0.012
path dependence	0.267	0.266
alliance	0.002	−0.001
log GDP (i)	0.027	0.106
log GDP (j)	0.007	−0.024
polity difference (ij)	−0.002	−0.003
log military expenditure (i)	0.052	0.005
log military expenditure (j)	0.0004	0.020
log population (i)	−0.042	−0.201
log population (j)	0.008	0.013
log distance (ij)	−0.009	−0.012
conflict (j)	0.001	0.001
embargo (j)	−0.0004	−0.002

The patterns changes in the post-Cold war period, illustrating that importers are now required to pay for arms transfers.

The results also suggest that only after the end of the Cold War have embargoes become effective in reducing the likelihood of arms transfers. Very striking are the estimates for our alliance dummy. Being allied has a positive, highly significant effect on the likelihood that *i* will export arms to *j*. However, this effect can only be found for the Cold War period. For the post-Cold War sample, the coefficient becomes negative. This effect is highly significant and captures the breakup of the Soviet bloc and a re-orientation of trade beyond the East-West divide.

To better gauge the effect magnitudes of our variables, we also calculated average marginal probability effects (AMPE)[16] for different quantiles (0, 0.25, 0.5, 0.75, 1) of a variable's distribution. The results presented in Table 3 show the AMPE when moving to a variable's maximum value.[17] The strongest effect for both periods can be found for 'path dependence'. Increasing previous export volumes to their maximum raises the trade probability by almost 27 percentage points. Interestingly, after the end of the Cold War, the effect of an exporter's GDP and population size increased markedly.

We also find sizeable effect magnitudes for some of our network measures. A past link increases, on average, the probability of future arms transfers by 7.7 (model 1) or 8.9 (model 4) points. By comparison, a conflict in country *j* increases the probability by 0.1 points. The effect of the number of export links is strongest for the post-Cold War period with an AMPE of 0.084.

Turning to the hyper-dyadic variables, we present partial dependence plots to visualize their effects. As Fig. 2 shows, 'export transitivity' increases trade probability by up to 2.7 (1955–1991) and 8.5 points (1992–2018), while Fig. 3 shows that shared supplier relationships exert an effect of 3.1 and 2.7 points, respectively. These effects are stronger than those of other, often cited drivers of trade like spatial distance, political distance, alliance membership or an importer's GDP and military expenditures.

The other significant network measures show smaller effect sizes, with, for instance, 'reciprocity' increasing trade probability by 0.5 or 0.2 points. Overall, these effect sizes make clear that network variables have, in relative terms, become more important after the end of the Cold War. Taking their absolute values together, the network measures change the predicted probability of a trade link by roughly 17 points for the sample period up until 1991 and by about 30 percentage points for the period after 1991.

5.2. Second stage estimation results

Table 4 presents our estimation results for the second stage of the Heckman model. Note that for all four models we can clearly reject the null that $\rho = 0$, thus confirming the necessity of using a sample selection model. Furthermore, there are 537474 (588754) observations in the Cold war (post-Cold War) sample in the selection equation, but only 9399 (9160) observations in the outcome equation. This shows that the vast majority of observations (528075 and 579594, respectively) are censored, which reflects the fact

[16] For numeric variables this is defined as $AMPE_{j,c} = \frac{1}{N}\sum_{i=1}^{N}(\varphi(x_i^T\beta_j)\beta_j)$ where φ is the density of standard normal distribution. For our binary variables we calculated $AMPE_{j,b} = \frac{1}{N}\sum_{i=1}^{N}(\Phi(x_{i,j=1}^T\beta) - \Phi(x_{i,j=0}^T\beta))$, where Φ is the CDF of a standard normal distribution and $x_{i,j=0}$ is set to 1 or 0 for variable j.

[17] The AMPEs for all quantiles can be found in the Online Appendix.

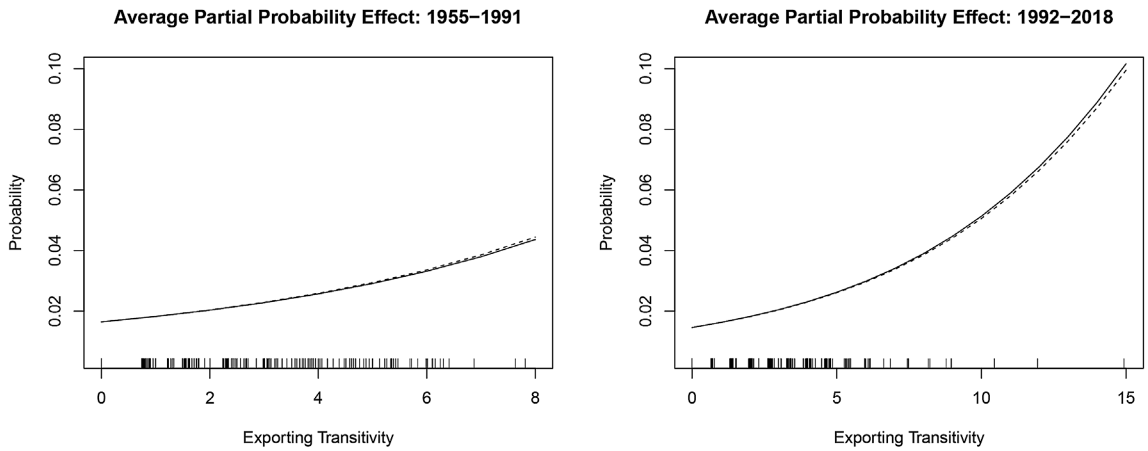


Fig. 2. Partial dependence plots, exporting transitivity.

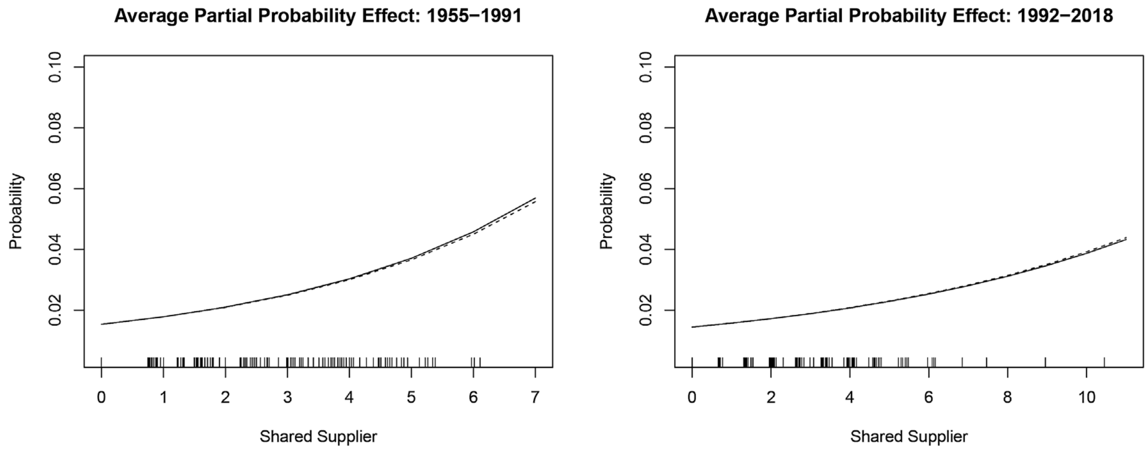


Fig. 3. Partial dependence plots, shared supplier.

that arms flows are a rare event.[18]

Looking at our network variables, we find that many of the coefficients that have been highly significant at the selection stage are no longer significant or robust. This is most notable, for the variables 'past link', 'exporting transitivity', 'shared customer' and 'shared supplier', although the latter remains positive and significant for the post-Cold War period. This confirms our expectation that cost reduction and information dissemination are less important for the determination of trade volumes. This is consistent with the notion that links costs are fixed and not related to the size of arms transfers.

The effect of reciprocity is strongly positive again, however. Having imported from *j* in the past increases the volumes of exports from *i* to *j* by about 250.00 TIV for the Cold War sample (model 2) and by 180.000 TIV for the post-Cold War sample (model 4). The number of exporting links of *i*, which according to the first-stage results increases trade probability, turns negative for the Cold War and is insignificant for the period after 1991. This, again, suggests that cost reduction and information dissemination are of less importance for trade volumes.

The fact that the coefficients for importer *j*'s number of export links and *i*'s number of import links remain negative and statistically significant is no surprise. They simply capture the fact that countries that export arms to a lot of other states have the capacity to produce arms and are therefore less reliant on weapon imports, while countries with many import ties are unlikely to be producers and thus less likely to export.

Notably, the coefficient for the number of import links of importer *j* is now positive and significant at a 5% level. In light of our theoretical approach this could mean that a large number of ties reduces the costs of and informational barriers to establishing new trade links, thus increasing trade volumes. However, we believe this rather captures the intensity of demand for arms imports. A

[18] Note that we also estimated the second stage as a single linear equation. While the results were not exceedingly different, a number of coefficients, most notably the now statistically significant (and counterintuitive) negative effect for 'past link' suggested that selection bias affected these estimations. The results can be found in the Online Appendix.

Table 4
Second stage results.

	(1) 1955–1991 Time Poly.	(2) 1955–1991 Time FE	(3) 1992–2018 Time Poly.	(4) 1992–2018 Time FE
Network Statistics				
past link	0.07	0.07	0.12	0.12
	(0.06)	(0.06)	(0.07)	(0.07)
reciprocity	0.24***	0.25***	0.17*	0.18*
	(0.07)	(0.07)	(0.07)	(0.07)
# exporting links (i)	−0.01***	−0.01***	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)
# exporting links (j)	−0.03***	−0.03***	−0.01***	−0.01***
	(0.00)	(0.00)	(0.00)	(0.00)
# importing links (i)	−0.05***	−0.05***	−0.06***	−0.06***
	(0.01)	(0.01)	(0.01)	(0.01)
# importing links (j)	0.04**	0.04**	0.03*	0.02*
	(0.01)	(0.01)	(0.01)	(0.01)
exporting transitivity	−0.01	−0.01	−0.03	−0.04*
	(0.02)	(0.02)	(0.02)	(0.02)
importing cycle	0.05	0.05	−0.02	−0.02
	(0.05)	(0.05)	(0.03)	(0.03)
shared supplier	0.01	0.01	0.10***	0.10***
	(0.03)	(0.03)	(0.02)	(0.02)
shared customer	0.02	0.02	0.01	0.01
	(0.01)	(0.01)	(0.01)	(0.01)
Controls				
path dependence	0.67***	0.67***	0.62***	0.62***
	(0.01)	(0.01)	(0.01)	(0.01)
alliance	0.00	0.01	0.10**	0.10**
	(0.04)	(0.04)	(0.04)	(0.04)
log GDP (i)	−0.08	−0.07	−0.00	0.00
	(0.04)	(0.04)	(0.04)	(0.04)
log GDP (j)	0.07**	0.08**	−0.05	−0.06*
	(0.02)	(0.02)	(0.03)	(0.03)
polity difference (ij)	−0.00	−0.00	0.01	0.00
	(0.00)	(0.00)	(0.00)	(0.00)
log military expenditure (i)	0.20***	0.20***	0.03	0.03
	(0.03)	(0.03)	(0.03)	(0.03)
log military expenditure (j)	0.15***	0.15***	0.18***	0.18***
	(0.02)	(0.02)	(0.02)	(0.02)
log population (i)	0.18***	0.18***	0.12***	0.11***
	(0.03)	(0.03)	(0.03)	(0.03)
log population (j)	−0.01	−0.01	0.04*	0.05**
	(0.02)	(0.02)	(0.02)	(0.02)
log distance (ij)	−0.01	−0.01	0.03	0.03
	(0.03)	(0.03)	(0.03)	(0.03)
embargo (j)	−0.34**	−0.37**	−0.09	−0.09
	(0.12)	(0.12)	(0.08)	(0.08)
ρ	0.30***	0.31***	0.30***	0.31***
	(0.03)	(0.03)	(0.03)	(0.03)
N	9399	9399	9160	9160

Note: i is the exporting country and j the importing country. Standard errors in parentheses. Coefficients of time dummies not shown. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

high number of trade partners is a reflection of a great demand for weapons.

Turning briefly to our controls, we find that the gravity variables lose their explanatory power. Spatial and political distances, as well as the exporter's GDP, are no longer significant. Hence, while gravity explains the creation of trade links, it does not influence trade volumes. Instead, demand factors, which are captured by an importer's military spending, are one of the most important factors explaining the size of transfers. Interestingly, being allied with each other had no effect on trade volumes during the Cold War, but a statistically significant, positive effect for the post-Cold War period. In other words, while allies are no longer necessarily preferred trading partners, they receive higher volumes. This suggest that alliance partners receive more advanced weaponry since more advanced arms have higher TIV values. Interestingly, in a reversal of the findings of the first-stage, our embargo variable is only significant and negative for the Cold War era but insignificant for the period after 1991. This implies that while arms embargoes did not significantly reduce the likelihood of exports during 1955–1991, they at least reduced trade volumes. The fact that we find no significant effect for the post-Cold War era is explained by the effectiveness of embargoes at the selection stage.

Finally, to convey a rough idea about relative effects sizes, we present in Table 5 the linear effect when a variable increases to its maximum value. Unsurprisingly, previous export magnitudes, an exporter's population size and an importer's military expenditures

Table 5
Linear effect when changing variable to its maximum value, second stage.

	(1) 1955–1991 Time Poly.	(l) 1992–2018 Time Poly.
past link	0.073	0.117
reciprocity	0.243	0.173
# exporting links (i)	−0.805	0.293
# exporting links (j)	−1.722	−0.813
# importing links (i)	−0.459	−1.171
# importing links (j)	0.390	0.488
exporting transitivity	−0.088	−0.515
importing cycle	0.368	−0.267
shared supplier	0.047	1.053
shared customer	0.413	0.201
path dependence	5.440	4.923
alliance	0.003	0.099
log GDP (i)	−2.236	−0.065
log GDP (j)	2.210	−1.654
polity difference (ij)	−0.084	0.107
log military expenditure (i)	2.658	0.383
log military expenditure (j)	2.026	2.428
log population (i)	3.653	2.468
log population (j)	−0.263	0.927
log distance (ij)	−0.044	0.086
embargo (j)	−0.341	−0.088

have the largest effect on export volumes. ‘Shared supplier’ has the strongest positive effect of our network variables, indicating that an increase to its maximum value increases exports by over a million TIV.

In sum, the second stage results suggest that the trade enhancing effect of the network variables is less pronounced with respect to trade volumes. Rather supply restrictions and demand intensity play a much greater role. Demand is mainly captured by the strong, positive effect of an importer’s military expenditures which is highly significant across all specifications.

5.3. Comparing model performances

Our central theoretical claim is that network structures strongly affect the evolution of the IAT. While the parameter estimates of our Heckman models seem to confirm these predictions, we still need to analyze the relative performance of estimation models with and without network variables.[19] This can be done by conducting in-sample and out-of-sample predictions for both estimations stages. This allows us to compare model performance on a yearly basis for the whole sample period. Given that they are more demanding, we focus here on the out-of-sample predictions using ‘Area Under Curve” (AUC) measures.[20] Given our one-year lag structure, we decided to predict yearly trade links and trade volumes based on the fitted model for the previous year.

For the first estimation stage we use precision recall curves (PR AUC). Alternatively, we could also employ ROC-curves (‘Receiver Operating Characteristic’). However, arms trading is a rare event and PR AUC puts greater weight on true positive predictions. We therefore follow the recommendation by Cranmer and Desmarais (2017) and Thurner et al. (2019) that PR AUC should be preferred for rare-event binary classes.[21] For the second stage, we evaluate relative model performances using the mean squared residuals (MSR). Given the selection equation, these are obtained as follows:

$$MSR_t = n_t^{-1} \sum_{\{ij:D_{ij}=1\}}^{n_t} (Y_{ij,t} - E[Y_{ij,t}|X_{ij,t-1}, D_{ij,t} = 1])^2 \tag{6}$$

Fig. 4 presents the out-of-sample predictive performance for both stages of a model with network variables (black line) versus without network variables (dashed line).[22] Note that a model with a greater AUC or smaller MSR performs better. The left panel presents the results for the first stage. We find that the model with network statistics outperforms the other model for the whole sample period. The difference is smallest for the end of the 1960s but gets larger afterwards and remains relatively stable between 1992 and 2018. Note also that the predictive power of both models clearly declines with the end of the Cold War but increases again somewhat after 2000. In terms of the relative importance of our network dependency measures, we find that ‘past link’, ‘exporting transitivity’, ‘shared supplier’ and ‘# exporting links (i) have the strongest predictive impact.

[19] Note that when estimating our models without network variables, we find that the results for the control variables do not change much (see Online Appendix). Hence, inclusion of network factors adds additional explanatory power without changing the impact of these other drivers of the IAT.

[20] The results of the in-sample predictions can be found in the Online Appendix.

[21] The in-sample and out-of-sample predictions using ROC-AUCs can be found in the Online Appendix.

[22] The in-sample results can be found in the Online Appendix.

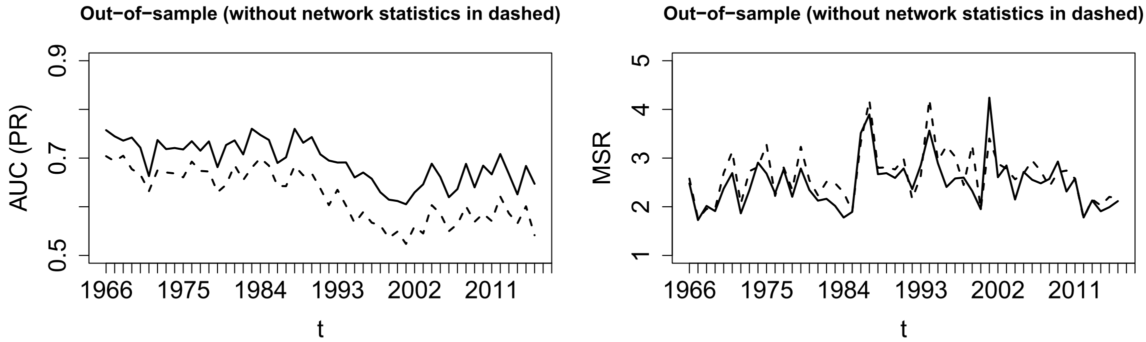


Fig. 4. Predictive performance, out-of-sample. Black: with network variables, Dashed: without network variables.

The right panel of the figure shows the results for the second stage and clearly demonstrates that the difference in model performances is much smaller than compared to the first stage. Furthermore, while the model with network variables still slightly outperforms the model without these variables, there are a number of years (i.e. beginning 2000s) where the latter actually provides more accurate predictions.

In sum, our out-of-sample predictions corroborate that models that control for network structures are clearly better at explaining and predicting arms trade patterns than models that ignore these factors. While this finding holds for both the extensive and the intensive margin, our analysis also clearly shows that network properties are more important for establishing trade links (i.e., the selection of export destinations) than for the determination of trade volumes.

6. Summary and conclusion

This paper contributes in two important ways to the growing literature that analyzes the international arms trade from a network perspective. First, it proposes a theoretical account of the mechanisms that relate network structures with countries' arms trading behavior. Second, it simultaneously estimates the effects of network structures on both the intensive and extensive margins of the international weapons trade.

We argue that the costs of establishing trade links and incomplete information about these costs are an essential impediment to trade. We show that existing network structures, i.e. dyadic and hyper-dyadic trade relationships, reduce link costs and informational barriers. Costs can be due to economic, political or security factors. We argue that cost reduction and information dissemination are vital in understanding the evolution of the arms trade network. Therefore, the IAT is not only driven by exogenous economic and security factors but also by endogenous network processes.

We test our empirical expectations by estimating gravity equations in the framework of a network analysis that employs a two-stage selection model. To our knowledge, we are the first to combine a Heckman model with a network analysis to estimate the effects of network properties on both the selection of trade partners and the trade volumes. Our specifications, which include dyadic and hyper-dyadic relationship variables, are applied to both the Cold War period (1955–1991) and the post-Cold War era (1992–2018). Our results show that most of the network variables have the expected effect on the selection stage. This suggests, at the very least, that link costs and informational barriers do shape countries' choices of trading partners. We find that network structures are less relevant for explaining trade volumes. Levels of arms transfers are much more determined by supply and demand factors, with the latter being expressed by an importer's military spending. This also indicates that link costs are fixed and not proportional to trade volumes. We used out-of-sample predictions to evaluate the relative performance of models with and without network variables. We find the former to clearly outperform the latter – especially with respect to the selection stage.

On a general level, our findings support previous research that emphasizes the importance of network centrality for the way countries make security-relevant choices. For instance, Bove et al. (2016) show that a states' central position in the global oil network affects its domestic military spending. Kinne (2012) finds that trade integration affects conflict behavior. The higher the number of direct and indirect trade ties as well as their values, the less likely is a country to initiate military interstate conflicts. In a similar vein, our analysis clearly shows that existing network structures affect future arms export choices. A country's position in the arms network as measured by its dyadic and hyper-dyadic relationships shapes the international flow of weapons.

On a technical level, our results clearly suggest that future research on the IAT should take the effect of network structures into account. This dictum holds even if one is solely interested in the analysis of the intensive margin because ignoring selection into trade will bias estimates when analyzing trade volumes. On an empirical level, our results lead us to predict continuous future growth in the IAT. While global political or economic shocks may lead to a temporary reversal or slowing down, the long-run trend is upward. Globalization and the end of the bipolar world order have led to a growth in network density. Our research suggests that more links reduce link costs and help disseminate information which makes additional trade links more likely. On a political level, our analysis shows that political and security costs and the availability of information about them are importers drivers. Actors interested in stronger arms control and a reduction in the IAT should therefore aim at increasing the political costs for exporting governments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code and data files required to reproduce all the findings in this article are available at https://data.mendeley.com/datasets/xkxyrk5my8/1.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejpoleco.2021.102033.

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